Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

Section 1: Plane Shapes and Their Properties

A plane shape is a flat, two-dimensional (2-D) shape. It is called two-dimensional because it has only length and width — it has no thickness. Plane shapes include triangles, squares, rectangles, parallelograms, trapeziums, circles and many other polygons. In this section you will learn to name these shapes, state their properties, and notice their similarities and differences.

A triangle is a closed shape with three straight sides and three angles. Triangles come in three types, and you can tell which is which by looking at their sides and angles (see fig-01).

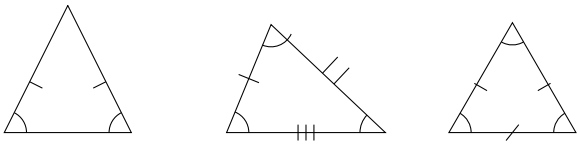
An isosceles triangle has two adjacent sides equal, the two base angles equal, and one line of symmetry.

A scalene triangle has no two sides equal, no two angles equal, and no line of symmetry.

An equilateral triangle has all three sides equal, all three angles equal (each is 60°), and three lines of symmetry.

So the three types are similar because they all have three sides and three angles, but they differ in how many sides and angles are equal.

Visual



Isosceles Scalene Equilateral

Isosceles: a single tick on each of the two slanted sides, an arc at each base angle. Scalene: one, two and three ticks on the three sides, an arc at each vertex. Equilateral: a single tick on each of the three sides, an arc at each vertex.

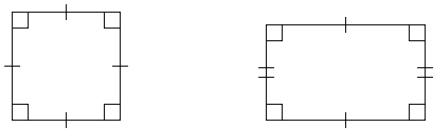
A square and a rectangle are both four-sided shapes whose angles are all equal (each is 90°) and that both have two pairs of parallel sides (see fig-02). Their difference is in the sides.

A square has all four sides equal, four lines of symmetry, and its diagonals bisect each other at a right angle.

A rectangle has two opposite sides equal (two longer sides and two shorter sides), two lines of symmetry, and its diagonals bisect each other but not at a right angle.

The two shapes are similar because both have all right angles and two pairs of parallel sides. They differ because the square's four sides are all equal, while the rectangle's opposite sides are equal.

Visual



Squares Rectangles

Square: a right-angle mark at each corner, a single tick on each of the four sides. Rectangle: a right-angle mark at each corner, a single tick on each of the two longer sides and a double tick on each of the two shorter sides.

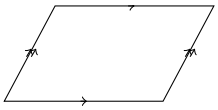
A parallelogram and a trapezium are both four-sided shapes, but they differ in how many sides are parallel (see fig-03).

A parallelogram has two pairs of opposite sides equal and parallel, its opposite angles are equal, it has no line of symmetry, and its diagonals divide it into two equal triangles.

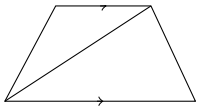
A trapezium has only one pair of parallel sides, its opposite angles are not equal, it has no line of symmetry, and a diagonal divides it into two unequal triangles.

So both shapes have no line of symmetry, but the parallelogram has two pairs of parallel sides while the trapezium has only one.

Visual



Parallelogram



Trapezium

Parallelogram: a single arrow on each of the two horizontal sides, a double arrow on each of the two slanted sides. Trapezium: a single arrow on each of the two parallel sides, and one diagonal drawn in.

A polygon is a closed shape that has three or more straight sides. Polygons are named by their number of sides. You can also work out how many triangles a polygon can be divided into: it is always two fewer than the number of sides, that is (number of sides) − 2.

Name of polygon	Number of sides	Number of triangles (n − 2)
Triangle	3	1
Quadrilateral	4	2
Pentagon	5	3
Hexagon	6	4
Heptagon	7	5
Octagon	8	6
Nonagon	9	7
Decagon	10	8
Hendecagon (undecagon)	11	9
Dodecagon	12	10

Use the table to name any polygon from its number of sides, and use the rule (sides − 2) to find how many triangles it can be divided into.

Example 1. Identify each shape below and state one property of each.

(a) A shape with all three sides equal.
Answer: This is an equilateral triangle. Its three angles are all equal (60° each) and it has three lines of symmetry.

(b) A four-sided shape with two pairs of parallel sides and all angles 90°, whose four sides are not all equal.
Answer: This is a rectangle. It has two lines of symmetry and its diagonals bisect each other but not at a right angle.

Example 2. How many triangles can a dodecagon (12 sides) be divided into?

Answer: Use the rule: number of triangles = number of sides − 2 = 12 − 2 = 10.

So a dodecagon can be divided into 10 triangles.

Practice

1. Name the type of triangle that has no two sides equal and no line of symmetry, and state one property of it.

2. A shape has all four sides equal, all angles 90°, and four lines of symmetry. What is it, and how does it differ from a rectangle?
3. State one similarity and one difference between a parallelogram and a trapezium.
4. How many triangles can an octagon be divided into if you use the rule (number of sides − 2)?
5. Which of these is a pentagon and which is a hexagon: a polygon with 5 sides, or a polygon with 6 sides?

Section 2: Parts of a Circle

A circle is the set of all points that are the same distance from a fixed point at its middle. That fixed point is the centre of the circle. Look at the two circles in fig-04 as you read the names below — every part is labelled on them.

The circumference is the whole curved boundary of the circle. Think of it as the perimeter of the circle — the distance all the way around the edge.

A radius is any straight line that joins the centre of the circle to the circumference. In fig-04 a radius is drawn from the centre down to the edge. Because every point on the circumference is the same distance from the centre, all the radii of one circle are equal in length.

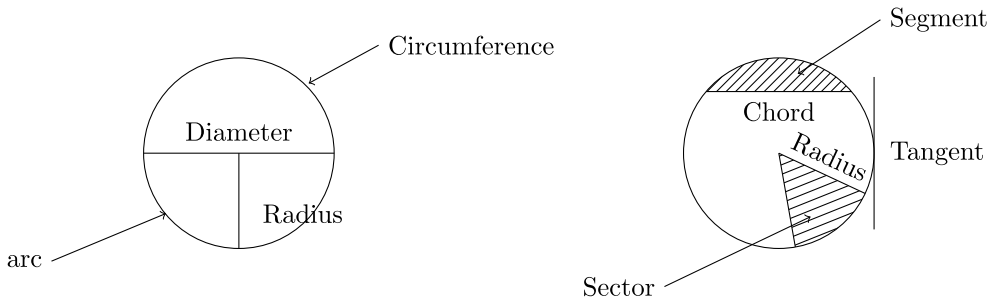
A diameter is any straight line that passes right across the circle, through the centre, from one point on the circumference to another. It is the longest straight line you can draw across a circle, and it is twice as long as a radius.

A chord is any straight line joining two points on the circumference. A chord does not have to pass through the centre — only a chord that passes through the centre is called a diameter. So every diameter is a chord, but a chord is a diameter only when it crosses the centre.

An arc is any curved part of the circumference. A small piece of the circumference is an arc, just as a section of a straight line is part of that line.

The remaining names pick out parts of the circle's inside and outside:
- A segment is the region bounded by a chord and the arc it cuts off — in fig-04 the part above the chord is shaded.
- A sector is the region bounded by two radii and the arc between them, like a slice of the circle. In fig-04 the shaded slice below is a sector.
- A tangent is a straight line that touches the circle at exactly one point and does not cut across it. In fig-04 it is the line touching the right-hand side of the second circle.

Visual



Two labelled circles. The first has the diameter drawn across it and a radius from its centre to the circumference, with the arc and the circumference labelled. The second has a chord near the top with the segment above it shaded, a shaded sector below with its right-hand bounding line labelled Radius, and a tangent touching it on the right.

A circle has a radius of 4 cm. Answer these about the same circle.

a) What is the length of its diameter?
A diameter passes through the centre and reaches the circumference on both sides, so it is twice the radius. 4 cm × 2 = 8 cm.

b) Is the diameter a chord? Is it an arc?
The diameter is a straight line joining two points on the circumference, so yes, it is a chord. It is not curved, so it is not an arc — an arc is a curved part of the circumference.

c) A straight line touches this circle at exactly one point. What is it called?
A line that touches a circle at exactly one point without crossing it is a tangent.

Practice

1. Name the part of the circle that is (a) any straight line joining the centre to the circumference, and (b) the curved part of the circumference.
2. A circle has a diameter of 10 cm. What is the length of its radius?
3. Explain in your own words: is every diameter a chord? Is every chord a diameter?
4. A straight line touches a circle at exactly one point. What is this line called? Name the region bounded by two radii and the arc between them.

Section 3: Finding the Perimeter of Plane Shapes

The **perimeter** of a shape is the total distance around it — you simply walk round the outside and add up every side you cross. In this section you will find the perimeter of a regular polygon, a square, a rectangle, a trapezium, a parallelogram and a circle, and you will also learn to work backwards to find a missing side or radius when the perimeter is already known.

A **square** has four equal sides. If each side is L, then its perimeter is

$$L + L + L + L = 4L.$$

A **rectangle** has two long sides (length L) and two short sides (breadth b), so its perimeter is

$$L + b + L + b = 2L + 2b = 2(L + b).$$

A **regular** polygon has all sides equal *and* all angles equal. (Equal sides alone are not enough — a rhombus has four equal sides but its angles are not all equal, so it is not regular.) If a regular polygon has n sides, each of length s, its perimeter is $n \times s$.

Worked example — square. A square has sides of 15 cm. Its perimeter is

$$L + L + L + L = 4L = 4 \times 15 = 60 \text{ cm}.$$

Worked example — rectangle. A rectangle has length 10 cm and breadth 7 cm. Its perimeter is

$$2L + 2b = 2(10) + 2(7) = 2(17) = 34 \text{ cm}.$$

Worked example — finding a missing breadth. The perimeter of a rectangle is 60 cm and its length is 17 cm. What is its breadth?

Use $P = 2L + 2b$ with $P = 60$ and $L = 17$:

$$60 = 2(17) + 2b = 34 + 2b.$$

Collect like terms:

$$60 - 34 = 2b \quad \Rightarrow \quad 26 = 2b.$$

Divide both sides by 2:

$$b = \tfrac{26}{2} = 13 \text{ cm}.$$

So the breadth is **13 cm**.

Worked example — trapezium. A trapezium has four sides. Just add all four lengths. For a trapezium with sides 11 cm, 9 cm, 5 cm and 4 cm:

$$11 + 9 + 5 + 4 = 29 \text{ cm}.$$

(There is no special formula here — different trapezia have different side lengths, so the perimeter is always found by adding the four sides you are given.)

Worked example — parallelogram. A parallelogram has two pairs of equal sides. If the base a is 9 cm and the slanted side b is 5 cm:

$$a + a + b + b = 2a + 2b = 2(a + b) = 2(9 + 5) = 2(14) = 28 \text{ cm}.$$

Worked example — regular hexagon. A regular hexagon has 6 equal sides. If each side is 5.2 cm:

$$\text{perimeter} = 6 \times 5.2 = 31.2 \text{ cm}.$$

The perimeter of a circle is called its **circumference**. It is given by

$$\text{circumference} = 2\pi r \quad \text{or} \quad \pi d,$$

where r is the radius and d is the diameter (twice the radius). You are usually told which value of π to use, such as $\pi = \frac{22}{7}$ or $\pi = 3.14$.

Worked example — circumference. A circle has radius 7 cm. Using $\pi = \frac{22}{7}$:

$$2\pi r = 2 \times \tfrac{22}{7} \times 7 = 2 \times 22 = 44 \text{ cm}.$$

You could get the same answer through the diameter: $d = 2 \times 7 = 14$ cm, so

$$\pi d = \tfrac{22}{7} \times 14 = 22 \times 2 = 44 \text{ cm}.$$

Worked example — finding the radius from the circumference. The circumference of a circle is 440 cm. Find its radius, using $\pi = \frac{22}{7}$.

$$\text{circumference} = 2\pi r \quad \Rightarrow \quad 440 = 2 \times \tfrac{22}{7} \times r = \tfrac{44r}{7}.$$

Cross-multiply:

$$440 \times 7 = 44r \quad \Rightarrow \quad 3080 = 44r.$$

Divide both sides by 44:

$$r = \tfrac{3080}{44} = 70 \text{ cm}.$$

Worked example — a partial shape. A shape is a quarter circle: two straight sides of 4.9 cm meeting at a right angle, closed by a curved arc. Its perimeter is the two straight sides plus the curved part.

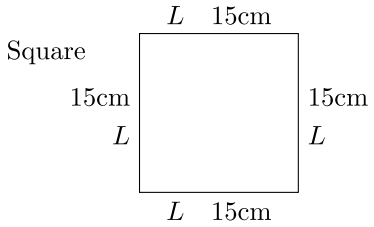
The curved part is one quarter of a full circle:

$$\tfrac{1}{4} \times 2\pi r = \tfrac{1}{4} \times 2 \times \tfrac{22}{7} \times 4.9 = 7.7 \text{ cm}.$$

So the perimeter is

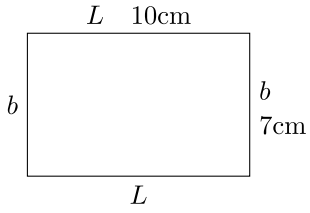
$$4.9 + 4.9 + 7.7 = 17.5 \text{ cm}.$$

Visual



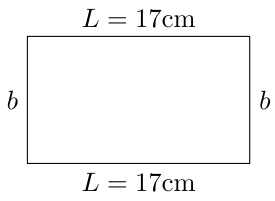
A square, named *Square* at its left, with all four sides marked twice — L and 15cm.

Visual



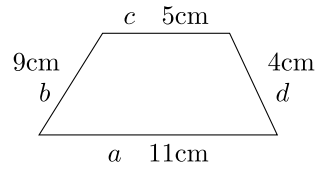
A rectangle with both long sides marked L, the top one also 10cm, and both short sides marked b, the right-hand one also 7cm.

Visual



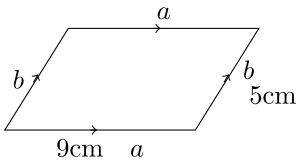
A rectangle with both long sides marked $L = 17$cm and both short sides marked b.

Visual



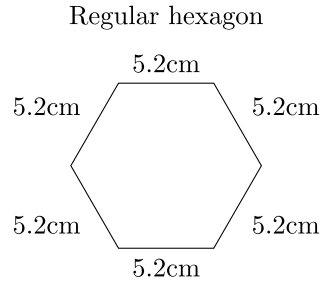
A trapezium with each side lettered and measured: the top c, 5cm; the left b, 9cm; the right d, 4cm; the base a, 11cm.

Visual



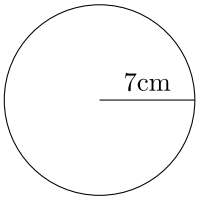
A parallelogram with a single arrow on each of its four sides: the two horizontal ones both lettered a, the lower also measured 9cm, and the two slanted ones both lettered b, the right-hand one also measured 5cm.

Visual



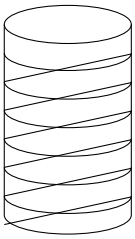
A regular hexagon, named above the drawing, with each of its six sides marked 5.2cm.

Visual



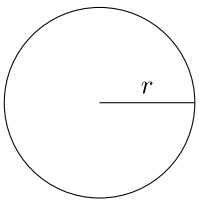
A circle with a radius drawn from the centre to the right of the circumference, marked 7cm.

Visual



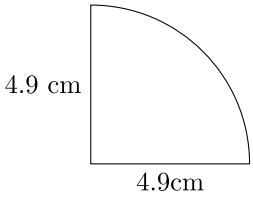
An upright cylinder with a rope wound round it, drawn as a series of turns spiralling down the body from the top face to the base.

Visual



A circle with a radius drawn from the centre to the right of the circumference, marked r.

Visual



A quarter circle: an upright side and a base meeting at a right angle, each marked 4.9cm, with an arc closing them from the top of the upright to the end of the base.

Practice

1. Find the perimeter of a square whose sides are 8 cm.
2. Find the perimeter of a rectangle of length 7 cm and breadth 4 cm.
3. What is the perimeter of a regular pentagon with sides 5 cm?
4. Find the perimeter of a regular hexagon with sides 5 cm.
5. Find the perimeter of a parallelogram whose base is 6 cm and sides are 5 cm.
6. A rectangle is such that its length is twice its breadth. If the breadth is d, write down the total length of all four sides in terms of d.
7. The circumference of a circle is 22 cm. Find the radius, using $\pi = \frac{22}{7}$.

Check your work

Attempt every question before you read this page.

Plane Shapes and Their Properties

1. Scalene triangle. No two sides are equal and no angle is equal to another.
2. It is a square. Unlike a rectangle, all four sides of a square are equal (a rectangle has only opposite sides equal), and the diagonals of a square bisect each other at a right angle.
3. Similarity: both have no line of symmetry. Difference: a parallelogram has two pairs of parallel sides, while a trapezium has only one pair of parallel sides.
4. 6 triangles. An octagon has 8 sides, so $8 - 2 = 6$.

   Use the rule: number of triangles = number of sides − 2. An octagon has 8 sides, so $8 - 2 = 6$.
5. A polygon with 5 sides is a pentagon (it divides into 3 triangles), and a polygon with 6 sides is a hexagon (it divides into 4 triangles).

Parts of a Circle

1. (a) Radius; (b) Arc.
2. 5 cm (the radius is half the diameter).
3. Every diameter is a chord, because it is a straight line joining two points on the circumference. But a chord is a diameter only when it passes through the centre, so not every chord is a diameter.
4. The line is a tangent; the region bounded by two radii and the arc between them is a sector.

Finding the Perimeter of Plane Shapes

1. 32 cm

   Perimeter of a square $= 4L = 4 \times 8 = 32$ cm.
2. 22 cm

   $2L + 2b = 2(7) + 2(4) = 14 + 8 = 22$ cm, or $2(7 + 4) = 2(11) = 22$ cm.
3. 25 cm

   A regular pentagon has 5 equal sides, so perimeter $= 5 \times 5 = 25$ cm.
4. 30 cm

   A regular hexagon has 6 equal sides, so perimeter $= 6 \times 5 = 30$ cm.
5. 22 cm

   $2(a + b) = 2(6 + 5) = 2(11) = 22$ cm.
6. $6d$

   Length $= 2d$, breadth $= d$. Perimeter $= 2(\text{length} + \text{breadth}) = 2(2d + d) = 2(3d) = 6d$.
7. 3.5 cm

   $2\pi r = 22$, so $2 \times \frac{22}{7} \times r = 22$, giving $\frac{44r}{7} = 22$. Cross-multiply: $44r = 154$, so $r = \frac{154}{44} = 3.5$ cm.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-077e033f	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____